

D1: Sensor Count and Layout

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What does each additional sensor buy, and where should it go?

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SHARC buoy project Code: `runD1SensorLayout.m`, `d1LayoutTools.m` Results: `Results/D1/D1_20261002_145943`

Summary

Question. For $n_s = 2, 3, 4$ sensors on the 33-point candidate grid, which layout and reference minimise the parameter uncertainty of the inverse, how much does each extra sensor buy, and how do the Level 2C and Gramian designs compare? **Outcome (LSM6DSV16X, p_0 , single wave direction).** **Two sensors suffice:** the best pair (s6 and reference s18) gives $\sigma_{\ln \beta} = 0.46\%$ and is Identifiable. A third sensor lowers $\sigma_{\ln \beta}$ by 19.6% and a fourth by 10.2%, both below the pre-registered 20% worth-it rule. All optima lie on the wave incidence axis and are nested. The reference choice barely matters (median $\times 1.00$). Level 2C is near-optimal (rank 167 of 40 920, efficiency $\eta = 0.89$); the Gramian $\lambda_{\min}$ designs are poor for this inverse ($\eta = 0.21$ to 0.35). **The real value of extra sensors is robustness**, not precision: at $n_s = 2$ only 4 of 528 layouts come within 10% of the optimum, at $n_s = 4$ 101 do. The optimum depends on knowing the wave direction, which is the subject of the follow-up study D3.

1 What the study does

Fisher information per layout. For a layout $\mathcal{L}$ with reference r , the features are the real and imaginary parts of $T_j = S_{jr}/(S_{rr} - S_{nn})$ for the $n_s - 1$ non-reference sensors, and

$$F(\mathcal{L}, r) = \sum_k J_k^T \Sigma_k^{-1} J_k, \quad \sigma_{\ln \beta}^2 = (F^{-1})_{11}, \quad \sigma_{\ln R}^2 = (F^{-1})_{22},$$

with Σ_k the full shared-reference covariance (complex Wishart, $N_{\text{eff}} - 1$ degrees of freedom) of the E1 engine. The field H and $\partial H / \partial \ln p$ are computed once at all 33 points; each layout extracts its rows, so the 181 104 (layout, reference) evaluations per noise case need no new EMM solves.

Criteria. Primary: A-optimal, minimise $A = \text{tr}(F^{-1}) = \sigma_{\ln \beta}^2 + \sigma_{\ln R}^2$. Secondary: D-optimal ($\log \det F$) and E-optimal ($\lambda_{\min}(F)$). Each criterion chooses its own best reference. Efficiency of a design: $\eta = A_{\text{opt}}/A_{\text{design}} \in (0, 1]$. Worth-it rule (pre-registered): sensor $n + 1$ is worth adding if $G = 1 - \sigma_{\ln \beta}(n + 1)/\sigma_{\ln \beta}(n) \geq 20\%$ at the optimum. Recommended count: the smallest n_s whose optimum is Identifiable at LSM6DSV16X noise.

Setting. p_0 ($R_{\text{phys}} = 0.72$ m), setting F ($N = 65536$, $L = 2048$, 84 bins in 3.178 to 8.301 rad/s), JONSWAP $H_s = 0.05$ m, single wave direction along the incidence axis. Grid: centre s1, then rings at $r/R = 0.3, 0.5, 0.7, 0.9$ (s2 to s9, s10 to s17, s18 to s25, s26 to s33) at eight angles $0, \pi/4, \dots, 7\pi/4$. Admissibility: $r \leq 0.9R$ and spacing ≥ 0.10 m; the closest grid points are 0.144 m apart, so every layout is admissible (528, 5456 and 40 920). Spacing sensitivity: 0.05 and 0.15 m. Noise: LSM6DSV16X (headline), ICM-42688-P and G3.

Checks. Gate: Level 2C with s26 matches `evaluateScenario` in all three noise cases (relative differences $\leq 4 \times 10^{-12}$); all 84 bins accepted and all blocks positive definite. The Gramian optima are recomputed exactly as in `level2C_finalCheck` (modes at 2.70 and 6.00 rad/s, $\Delta t = 0.02$ s, $N = 4001$). Unit tests 10/10.

2 Results

Optimal designs (LSM6DSV16X). Gain G is relative to the $n_s - 1$ optimum.

n_s	A-optimal layout	Ref	$\sigma_{\ln \beta}$	$\sigma_{\ln R}$	d_W	$\kappa(F)$	G	Class
2	s6, s18	s18	0.463%	0.110%	0.216	43	–	Identifiable
3	s6, s14, s18	s18	0.372%	0.086%	0.283	32	19.6%	Identifiable
4	s6, s14, s18, s26	s26	0.334%	0.070%	0.343	66	10.2%	Identifiable

n_s	D-optimal	$\sigma_{\ln \beta}$	E-optimal	$\sigma_{\ln \beta}$	Spacing 0.15 m (A)	$\sigma_{\ln \beta}$
2	s6, s26	0.512%	s6, s18	0.463%	unchanged	0.463%
3	s6, s26, s30	0.444%	s6, s14, s18	0.372%	s14, s15, s18	0.377%
4	s6, s7, s26, s30	0.375%	s6, s14, s18, s25	0.335%	s10, s14, s15, s26	0.344%

Spacing 0.05 m and unconstrained: identical to the 0.10 m optimum. Other noise cases: the same A-optimal layouts; $\sigma_{\ln \beta} = 0.635, 0.508, 0.455\%$ (ICM-42688-P) and $0.613, 0.491, 0.439\%$ (G3), all Identifiable, recommended $n_s = 2$.

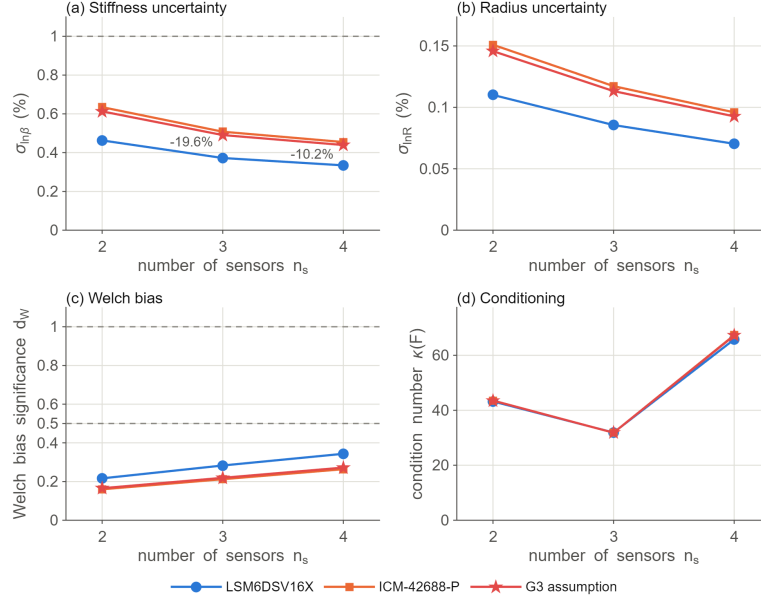


Figure 1. Best achievable (a) $\sigma_{\ln \beta}$, (b) $\sigma_{\ln R}$, (c) Welch bias significance d_W and (d) condition number $\kappa(F)$ against n_s , at the A-optimal layout, for three noise levels. Annotations in (a): gain per added sensor (LSM6DSV16X). Dashed lines: the 1% and $d = 0.5$, 1 class thresholds.

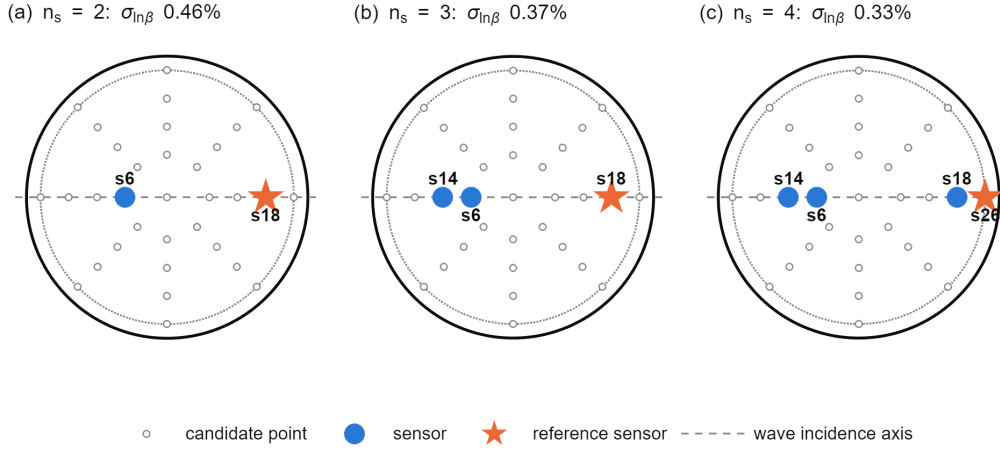
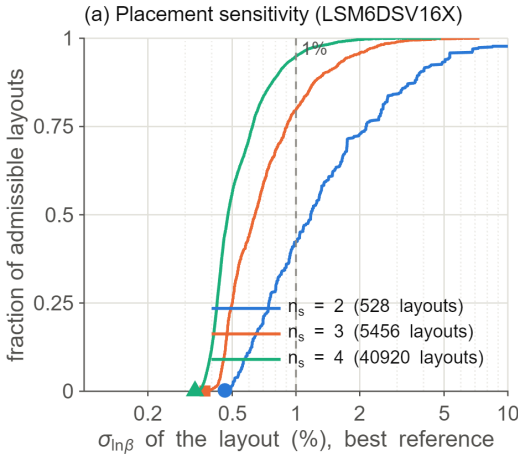


Figure 2. A-optimal layouts for $n_s = 2, 3, 4$ (LSM6DSV16X), viewed from above. Open circles: candidate points; dotted circle: $r = 0.9R$ admissibility limit; dashed line: wave incidence axis.

How forgiving is placement? Distribution of $\sigma_{\ln \beta}$ over all layouts, each with its best reference.



	$n_s = 2$	$n_s = 3$	$n_s = 4$
Layouts	528	5456	40 920
Optimum $\sigma_{\ln \beta}$	0.463%	0.372%	0.334%
Median $\sigma_{\ln \beta}$	1.174%	0.640%	0.478%
95th percentile	5.32%	1.86%	1.01%
Within 5% of optimum	1	3	11
Within 10% of optimum	4	13	101

Figure 3 (left). Empirical distribution of $\sigma_{\ln \beta}$ over all admissible layouts (log scale); markers: optima; dashed line: 1%. Roughly 40%, 80% and 95% of the layouts reach 1% for $n_s = 2, 3, 4$. Choosing the best instead of the worst reference changes A by a median factor of 1.00.

Comparison with earlier designs ($\eta = A_{\text{opt}}/A_{\text{design}}$; rank in the A ordering, best reference).

Design	Layout	Rank	η
Level 2C ($n_s = 4$)	s2, s6, s14, s26 (best reference s26, confirmed)	167 / 40 920	0.893
Gramian $\lambda_{\min}$, $n_s = 2$	s26, s30	228 / 528	0.212
Gramian $\lambda_{\min}$, $n_s = 3$	s26, s27, s30 (2 tied)	3587–3588 / 5456	0.236
Gramian $\lambda_{\min}$, $n_s = 4$	s26, s27, s29, s30 (4 tied)	27 186–27 189 / 40 920	0.354

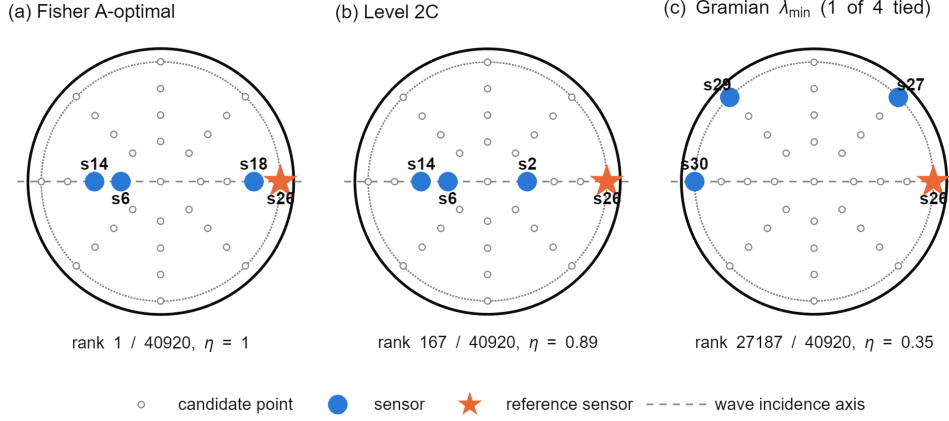


Figure 4. Four-sensor designs compared: (a) Fisher A-optimum, (b) Level 2C, (c) one of the four tied Gramian $\lambda_{\min}$ optima, with their rank and efficiency η in the Fisher A ordering (LSM6DSV16X).

3 Interpretation

1. **Two sensors are enough under the pre-registered rule.** The optimal pair is already Identifiable with a margin of two on $\sigma_{\ln \beta}$. The third sensor's 19.6% gain is just below the 20% threshold and the fourth's 10.2% clearly below: one transmissibility already gives two real features per bin for two parameters. The result holds at all three noise levels.
2. **Geometry of the optimum.** Every A-optimum lies on the incidence axis: one sensor at $0.3R$ (s6), the reference at $0.7R$ (s18) on the opposite side, then s14 at $0.5R$ next to s6, then s26 at the edge. The designs are nested, so a deployment can be extended without moving sensors. Plausibly, the phase of the response changes fastest along the axis, so on-axis pairs carry the most information on the flexural wavelength.
3. **Extra sensors buy robustness.** The optimum barely improves beyond $n_s = 2$, but the distribution tightens sharply: the median drops from 1.17% to 0.48% and the 95th percentile from 5.3% to 1.0%. With two sensors placement is critical (only 4 layouts within 10% of the optimum); with four almost any sensible layout works. The optimum assumes a known direction and exact positions; a deployment guarantees neither.
4. **The reference choice does not matter for variance.** With the full shared-reference covariance, changing the reference re-expresses the same cross-spectral information, so F is nearly invariant (median ratio 1.00). The reference can therefore be chosen on practical grounds, for example the highest-SNR point, which also relaxes the noise-floor requirement of R3.
5. **Level 2C is a good design for this inverse.** Rank 167 of 40 920 with $\eta = 0.89$: it shares s6, s14 and s26 with the four-sensor optimum and differs only in s2 against s18. The s26 reference is confirmed as the best reference for Level 2C.
6. **Modal observability is not parameter identifiability.** The Gramian $\lambda_{\min}$ designs place sensors on the edge ring, where the two modes are most observable, and are poor for the Fisher criterion ($\eta = 0.21$ to 0.35 , bottom third of the ranking). Each criterion suits its own method (Gramian for SSI, Fisher A for this inverse); the Gramian is not a usable proxy here.
7. **Criterion and spacing.** E-optimal designs coincide with the A-optimal ones for $n_s \leq 3$; D-optimal designs favour the edge and cost 11 to 19% in $\sigma_{\ln \beta}$: $\log \det F$ weights both parameter directions equally, so it can trade β precision for R precision. A 0.15 m spacing rule costs only 1 to 3%.
8. **Bias grows with n_s .** d_W rises from 0.22 to 0.34 as σ falls and the Welch bias does not (as in D2); it stays below 0.5.

Limitations.

- **Single, known wave direction.** The optima sit on the incidence axis because the forcing is unidirectional with a known heading. A real sea has directional spread or several components, which makes the spectral matrix rank two or higher and changes T itself. The rotation-robustness check printed by `plotD1SensorLayout` (efficiency of each optimum rotated in 45° steps) is the first measure of this; the full answer is D3, which repeats this search under the V2 directional forcing model.
- **One floe, one sea state.** p_0 and JONSWAP $H_s = 0.05$ m; A1 and R2 test other floes and spectra.
- **Grid, positions, linearisation.** 33 candidate points; position errors in V1; M1 checks the two-sensor optimum.

Requirement for the experiment and the thesis

Under a known, single wave direction, **two sensors on the incidence axis** (s6 at $0.3R$ and reference s18 at $0.7R$ on the opposite side) meet the Identifiable class with LSM6DSV16X, ICM-42688-P or G3-level noise. Because the two-sensor optimum is sensitive to placement and to the assumed direction, deploy **three or four** sensors in the nested sequence s6, s18, s14, s26 (or Level 2C, $\eta = 0.89$) until D3 has quantified the directional penalty. Choose the reference by SNR, not by variance. Do not use the Gramian criterion to place sensors for this inverse.

Outputs (Results/D1/): D1_20261002_145943.log/.mat/_summary.txt; D1_optimum.csv, D1_sensor_count.csv, D1_distribution.csv, D1_comparison.csv; figures/D1_fig1 to D1_fig4 (.pdf/.png).